电动力学导论

Introduction to Electrodynamics

Nan LIN
上海交通大学
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上海交通大學
SHANGHAI JIAO TONG UNIVERSITY

1 Electric Fields in Matter

Polarization
The field of a polarized object

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Dielectrics

In most everday objects belong to one of two larges classes:

- **Conductors**: charges aree free to move about through the material, not associated twith any particular nucleus but roam around at will
- **Dielectrics (Insulators)**: All charges are attached to specific atoms or molecules, all they can do is move a bit within the atom or molecule.

Two principal mechanisms: Electric fields can distort the charge distribution of a dielectric atom or molecule: *stretching* and *rotating*.

Induced Dipoles

What happens to a neutral atom when it is placed in electric field $\vec{E}$?

- Although the atom as a whole is electrically neutral, there is a positively charged core (the nucleus) and a negatively charged electron.
- The nucleus is pushed in the direction of the field, and the electrons the opposite way.

The atom is **polarized**, with a tiny **dipole moment** $\vec{p}$, which points in the same direction as $\vec{E}$, proportional to the field:

$$\vec{p} = \alpha \vec{E} \tag{1}$$

where α is called **atomic polarizability**.

Induced Dipoles

For some neutral molecules like carbon dioxide, they frequently polarize more readily in some directions than in others, for example:

$$\vec{p} = \alpha_{\perp}\vec{E}_{\perp} + \alpha_{\parallel}\vec{E}_{\parallel} \tag{2}$$

For a completely asymmetrical molecule, there is the most general linear relationship between $\vec{E}$ and $\vec{p}$:

$$\begin{cases} p_x = \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z \\ p_y = \alpha_{yx}E_x + \alpha_{yy}E_y + \alpha_{zz}E_z \\ p_z = \alpha_{zx}E_x + \alpha_{zy}E_y + \alpha_{zz}E_z \end{cases} \tag{3}$$

The induced dipole moment may not be in the same direction as $\vec{E}$.

Alignment of polar molecules

For some molecules which have built-in, permanent dipole moments, called **polar molecules**, e.g. water molecule,

- If the champ is uniform, the force on the positive end exactly cancels the force on the negative end. However, there will be a torique

$$\vec{N} = \vec{F}_- \vec{r}_- + \vec{F}_+ \vec{r}_+ \tag{4}$$

$$= \frac{\vec{d}}{2} \times q\vec{E} - \frac{\vec{d}}{2} \times (-q\vec{E}) = q\vec{d} \times \vec{E} \tag{5}$$

$$= q\vec{d} \times \vec{E} = \vec{p} \times \vec{E} \tag{6}$$

Alignment of molecules and polar molecules

- If the champ is not uniform, $\overrightarrow{F}_+$ does not balance $\overrightarrow{F}_+$, there will be a net force on the dipole:

$$\overrightarrow{F} = \overrightarrow{F}_+ + \overrightarrow{F}_- = q(\overrightarrow{E}_+ - \overrightarrow{E}_-) = q(\Delta\overrightarrow{E}) \tag{7}$$

  Assuming the dipole is very short, we approximate the small change in the electric field:

$$\Delta E_x = \nabla E_x . \overrightarrow{d}, \quad \Delta\overrightarrow{E} = (\overrightarrow{d}.\nabla)\overrightarrow{E} \tag{8}$$

  and therefore

$$\overrightarrow{F} = (\overrightarrow{p}.\nabla)\overrightarrow{E} \tag{9}$$

The torque about the center of the dipole in a nonuniform field is:

$$\overrightarrow{N} = \overrightarrow{p} \times \overrightarrow{E} + \overrightarrow{r} \times \overrightarrow{F} \tag{10}$$

Polarization - Conclusion

The result of placing a piece of dielectric material in an electric field is that the material becomes **polarized**, i.e. pointing along the direction of the field, either a tiny dipole moment is induced for neutral atoms, or a torque for polar molecules.
A measure for this effect is

$$\vec{P} = \text{dipole moment per unit volume} \tag{11}$$

which is called **polarization**.

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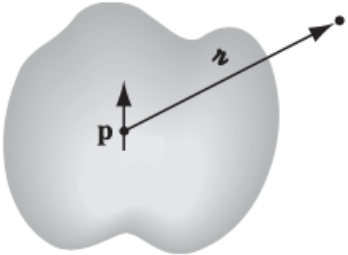
Bound Charges

Suppose we have a piece of polaried material, i.e. a lot of microscopic dipoles *lined up*. Given $\overrightarrow{P}(\overrightarrow{r})$, what is the field produced by this object (i.e. caused by the polarization)?



Bound Charges

It is easier to work with the potentiel.
For a single dipole $\overrightarrow{p}$,

$$V(\overrightarrow{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\overrightarrow{p}.\widehat{\overrightarrow{\underline{r}}}}{\underline{r}^2} \tag{12}$$

The total potential is the integration of each volume element, whose dipole moment $\overrightarrow{p} = \overrightarrow{P}\mathrm{d}\tau'$

$$V = \frac{1}{4\pi\epsilon_0}\int_V \frac{\overrightarrow{P}(\overrightarrow{r}').\widehat{\overrightarrow{\underline{r}}}}{\underline{r}^2}\mathrm{d}\tau' \tag{13}$$

$$= \frac{1}{4\pi\epsilon_0}\int_V \overrightarrow{P}.\nabla' \tag{14}$$

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